Wenjia He Ph.D. candidate

wenjia.he@kaust.edu.sa +966-568147905 13823409423 Male

China King Abdullah University of Science and Technology

Google Scholar 0000-0001-8161-4642

Profile

My research focuses on AI-guided protein engineering, with the goal of understanding and controlling the sequence–function relationship of proteins and developing generative and learning-based frameworks for designing functional biomolecules. My work is organized around three interconnected directions: (1) functional protein design through closed-loop AI–experiment optimization, (2) the development of generative and foundation models for protein design, and (3) machine learning approaches for protein function prediction. Together, these efforts seek to advance protein engineering from a largely empirical practice toward a principled, predictive, and programmable discipline.

Education

2024/01 – Present
Jeddah, Saudi Arabia

Ph.D. in Computer Science, *KAUST*

- Research focus: machine learning, protein engineering, bioinformatics.
- Affiliations: Center of Excellence for Smart Health (KCSH), Center of Excellence on Generative AI, and the SFB research group supervised by Prof. Xin Gao.

2022/08 – 2023/12
Jeddah, Saudi Arabia

Master of Engineering in Computer Science, *KAUST*

- GPA: 3.82/4.0.
- Relevant coursework: Data Analytics, Deep Learning in Visual Computing, Deep Generative Modeling, Algorithms in Bioinformatics, Biophysics, Genomics.

2018/09 – 2022/06
Jinan, China

Bachelor of Engineering in Software Engineering (Artificial Intelligence Division), *Shandong University (SDU)*

- GPA: 90.29/100. (Honors Bachelor).
- Relevant coursework: Machine Learning, Natural Language Processing, Recommendation System, Design and Analysis of Algorithms.

Representative Publications

2026/12

AI-guided evolution uncovers under-sampled high-flux states in NRT1.1 by relieving distributed kinetic traps, *Nature Communications (Under Revision)*

Wenjia He, Wancheng Liu, Bingxu Qi, Xiaopeng Xu, Min Zhou, Weihua Wu, Lu Zhang*, Xin Gao*, Yang Wang*

2026/12

Host-context fitness learning enables industrial enzyme optimization beyond natural evolutionary constraits, *ACS Catalysis (Under Revision)*

Wenjia He, Chenjie Feng, Maolin He, Jie Lin, Xiaopeng Xu, Chao Zha, Bin Xiao, Fei Zheng, Aixi Bai*, Xin Gao*

2026/12

Unifying Protein Sequence Design by Programming the Multi-modal Functional Space, *Nature Machine Intelligence (Under Revision)*

Junbo Yin, **Wenjia He (co-first author)**, Xin Gao*

2026/12

ConTP: Reshaping transporter functional space to resolve substrate specificity beyond evolutionary proximity, *Briefings in Bioinformatics (Under Revision)*

Chenjie Feng, **Wenjia He (co-first author)**, Junbo Yin, Xiaopeng Xu, Min Zhou, Yunchuan Wang, Mark Tester, Xin Gao*

First/Co-first Author Publications

2026/03 **SegDesign: A modular framework for controllable protein segment engineering,** *Protein Science*
Chenjie Feng, Junbo Yin, Chao Zha, Mohammed Saif, Xiaopeng Xu, Xin Gao*, **Wenjia He***

2025/02 **AggNet: Advancing protein aggregation analysis through deep learning and protein language model,** *Protein Science*
Wenjia He, Xiaopeng Xu, Haoyang Li, Juexiao Zhou, Xin Gao*

2022/01 **Accelerating bioactive peptide discovery via mutual information-based meta-learning,** *Briefings in bioinformatics*
Wenjia He, Yi Jiang, Junru Jin, Zhongshen Li, Jiaojiao Zhao, Balachandran Manavalan, Ran Su, Xin Gao*, Leyi Wei*

2021/12 **Learning embedding features based on multisense-scaled attention architecture to improve the predictive performance of anticancer peptides,** *Bioinformatics*
Wenjia He, Yu Wang, Lizhen Cui, Ran Su*, Leyi Wei*

2021/07 **Computational prediction and interpretation of cell-specific replication origin sites from multiple eukaryotes by exploiting stacking framework,** *Briefings in Bioinformatics*
Leyi Wei*, **Wenjia He**, Adeel Malik, Ran Su, Lizhen Cui, Balachandran Manavalan*

Co-authored Publications

2026/03 **ProteinMCP: An agentic AI framework for autonomous protein engineering,** *Protein Science*
Xiaopeng Xu*, Chenjie Feng, Chao Zha, **Wenjia He**, Maolin He, Bin Xiao, Xin Gao*

2025/05 **CFP-Gen: Combinatorial functional protein generation via diffusion language models,** *International Conference on Machine Learning 2025*
Junbo Yin, Chao Zha, **Wenjia He**, Chencheng Xu, Xin Gao*

2024/07 **SANTO: a coarse-to-fine alignment and stitching method for spatial omics,** *Nature Communications*
Haoyang Li, Yingxin Lin, **Wenjia He**, Wenkai Han, Xiaopeng Xu, Chencheng Xu, Elva Gao, Hongyu Zhao*, Xin Gao*

2024/06 **HELM-GPT: de novo macrocyclic peptide design using generative pre-trained transformer,** *Bioinformatics*
Xiaopeng Xu, Chencheng Xu, **Wenjia He**, Lesong Wei, Haoyang Li, Juexiao Zhou, Ruochi Zhang, Yu Wang, Yuanpeng Xiong, Xin Gao*

2024/11 **An AI agent for fully automated multi-omic analyses,** *Advanced Science*
Juexiao Zhou, Bin Zhang, Guowei Li, Xiuying Chen, Haoyang Li, Xiaopeng Xu, Siyuan Chen, **Wenjia He**, Chencheng Xu, Liwei Liu*, Xin Gao*

2023/11 **CoraL: interpretable contrastive meta-learning for the prediction of cancer-associated ncRNA-encoded small peptides,** *Briefings in Bioinformatics*
Zhongshen Li, Junru Jin, **Wenjia He**, Wentao Long, Haoqing Yu, Xin Gao, Kenta Nakai, Quan Zou*, Leyi Wei*

2023/10 **A unified method to revoke the private data of patients in intelligent healthcare with audit to forget,** *Nature Communications*
Juexiao Zhou, Haoyang Li, Xingyu Liao, Bin Zhang, **Wenjia He**, Zhongxiao Li, Longxi Zhou, Xin Gao*

2021/12 **iDNA-ABT: advanced deep learning model for detecting DNA methylation with adaptive features and transductive information maximization,** *Bioinformatics*
Yingying Yu, **Wenjia He**, Junru Jin, Guobao Xiao, Lizhen Cui, Rao Zeng*, Leyi Wei*

2021/11 **PSSP-MVIRT: peptide secondary structure prediction based on a multi-view deep learning architecture,** *Briefings in Bioinformatics*
Xiao Cao, **Wenjia He**, Zitan Chen, Yifan Li, Lesong Wei, Lizhen Cui, Ran Su*, Leyi Wei*

DigitalShadow: An Early Warning System of Coronary Artery Disease based on Facial Foundation Model, *npj Digital Medicine (Under Revision)*
Juexiao Zhou, Zhongyi Han, Mankun Xin, Xingwei He, Guotao Wang, Jiaoyan Song, Gongning Luo, **Wenjia He**, Xintong Li, Yuetan Chu, Juanwen Chen, Bo Wang, Xia Wu, Wenwen Duan, Zhixia Guo, Liyan Bai, Yilin Pan, Xuefei Bi, Lu Liu, Long Feng, Xin Gao*, Xiaonan He*

Projects

2025/05 – Present **Design of Terminal Deoxynucleotidyl Transferase,** *Joint Industry–Academia Project*
- Computational lead: Engineered TdT variants with improved dNTP selectivity and incorporation fidelity through iterative design-in-the-loop optimization, expanding patent-compliant design space for industrial translation.

2025/05 – Present **Design of Recombinant Human Hyaluronidase,** *Joint Industry–Academia Project*
- Computational co-lead: Engineered rHuPH20 variants with enhanced activity, thermostability, substrate specificity, and reduced immunogenicity through iterative design-in-the-loop optimization, identifying lead candidates beyond patent-constrained sequence space.

2025/01 – Present **High-resolution Functional Annotation of Membrane Transporters,** *Research Project*
- Computational Lead: Developed ConTP for high-resolution functional annotation of membrane transporters, with state-of-the-art performance and strong out-of-distribution robustness in substrate specificity prediction and TC family classification.

2024/10 – Present **Multimodal Generative Modeling of Protein Function,** *Research Project*
- Computational co-lead: Developed CFP-Gen and FuncGen, two novel frameworks that unify protein sequence design through programmable multimodal functional spaces.

2024/03 – 2025/12 **Engineering of Industrial Alkaline Protease,** *Joint Industry–Academia Project*
- Computational contributor: Supported the computational design of Savinase variants via AI-driven iterative mutation optimization, enhancing catalytic activity, thermostability, and washing performance.

2024/03 – Present **Multi-objective Engineering of Industrial Phytase,** *Joint Industry–Academia Project*
- Computational lead: Co-optimized phytase variants with enhanced catalytic activity and thermostability through AI-driven mutation design in iterative design–build–test cycles beyond patent-constrained sequence space.

2023/10 – Present **AI-guided Evolution and Mechanistic Study of Nitrate Transporters,** *Research Project*
- Comuputational lead**:** Designed transport activity–enhancing AtNRT1.1 variants through iterative ML-guided evolution, validated in vitro and in vivo.

2023/01 – 2023/05 **Co-optimisation of Protein Solubility and Stability,** *Course Project*
- Computational lead: Utilized ProtT5 and VAE to model an editable latent space of proteins, enabling the generation of variants with enhanced solubility and stability.

Work Experience

2022/06 – 2022/08 Beijing, China **AI Algorithm Internship,** *SyneronTech*
Offline Internship. Investigation on PPI, CPI, and PepPI mehtods. Reproduce typical algorithms for benchmark comparison, such as OnionNet-2, DeepPurpose, CAMP

2022/01 – 2022/06 Beijing, China **AI Technology Algorithm Internship,** *BioMap*
Offline Internship. Guided by Xin Gao and Le Song. Investigated protein aggregation and developed algorithms (AutoML) to predict amyloid formation in antibody light chains.

2021/11 – 2022/06 Jeddah, Saudi Arabia **Visitng Student Research Program,** *King Abdullah University of Science and Technology*
Research internship. Guided by Professor Xin Gao.

Awards

2025/10 **Dean's List Award KAUST 2025,** *King Abdullah University of Science and Technology*

2020/11 **The college's role model & Star of Youth League branch of the year,** *Shandong University*

2020/10 **Recognition Award in the Sixth "Internet+" college student Innovation and Entrepreneurship Competition,** *Shandong University*

2020/10 **Bronze Prize in 2020 SDU "Challenge Cup" Entrepreneurship Competition,** *Shandong University*

2020/10 **First Prize in the China Undergraduate Mathematical Contest in Modeling,** *CUMCM, CSIAM, Shandong Division*

2020/09 **Microsoft Student Club (MSC) "Excellent Club Leader - Technology" of the year,** *Microsoft Research Asia (MSRA)*

2020/04 **Honorable Mention Prize in the Mathematical Contest Modeling,** *MCM/ICM, COMAP*

Conference

2021/12 **The 6th CCF Bioinformatics Conference (CBC 2021),** *participation*

2021/07 **The 7th National Conference on Computational Biology and Bioinformatics (NCCBB 2021),** *participation*

2021/05 **The Bioinformatics and Intelligent Information Processing Conference (BIIP 2021),** *participation and poster*

2020/12 **The 5th CCF Bioinformatics Conference (CBC 2020),** *participation*

Skills

Machine Learning & AI ●●●●●
PyTorch, PyG, contrastive learning, generative modeling (VAE, diffusion, transformers)

Protein Analysis & Design ●●●●●
Protein language models (ESM), structure prediction (AF3), protein design (RFdiffusion, ProteinMPNN)

Bioinformatics ●●●●●
MSA analysis, RING analysis, molecular dynamics, clustering analysis (MMseqs2, TMalign)

Application Development ●●●○○
Java, JavaScript, Vue, SpringBoot; full-stack web development and WeChat Mini Program development

Languages

Cantonese ●●●●●
Native

Chinese ●●●●●
Native / Fluent

English ●●●○○
IELTS (6.5)